Where is the most common site of narrowing in patients with congenital coarctation of aorta?

Clinical Presentation

Congenital CoA may present shortly after birth or later in childhood. The timing depends largely on the severity:

- Those with the least severe lesions have no symptoms. Children are often diagnosed when they are found to have high blood pressure on screening exams.
- Neonates who present shortly after birth have the most severe lesions and present in **cardiogenic shock** because the left ventricle (LV) struggles to pump blood through the narrowed aorta. Common signs include tachypnea, pale skin, and absent femoral pulses.
- Patients who present in childhood have less severe lesions. They may initially be asymptomatic or have very subtle symptoms, which is why they go undiagnosed u e claudication and chest pain with exertion.

CoA is associated with several other CHDs including bicuspid aortic valve, ventricular septal defect, and patent ductus arteriosus. CoA is also present in 10% to 15% of patients with Turner syndrome.

CLINICAL CORRELATION

Turner syndrome is a chromosomal abnormality in which a female has only one X chromosome (45, XO) and presents with findings such as short stature, lymphatic lesions (ie, webbed neck, lymphedema in feet and hands), and a horseshoe kidney.

How does the severity of CoA differ between patients presenting shortly after birth and those presenting during childhood?

Pathophysiology

CoAs are classified by the location of the lesion as it relates to the **ductus arteriosus.** Recall that the ductus arteriosus is the fetal artery that connects the pulmonary trunk (PT) to the aorta (Ao); it allows fetal blood to bypass the nonfunctioning fetal lungs and move directly into the systemic circulation. The ductus arteriosus closes soon after birth.

Preductal coarctations (Figure 2, left) occur proximal to the attachment of the ductus arteriosus, whereas postductal coarctations (Figure 2, middle) occur distal to the attachment of the ductus arteriosus. Ductal coarctations (Figure 2, right), which are the most common, arise immediately adjacent to the ductus arteriosus attachment.

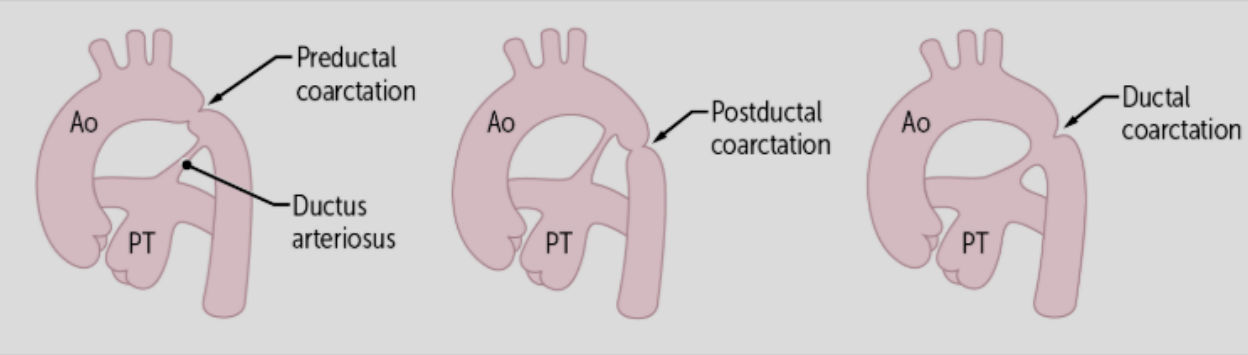


Figure 2 Types of coarctation of the aorta

Embryology. Congenital preductal CoA is likely due to the abnormal development of the aortic arch during formation from the embryonic left fourth aortic arch artery, while postductal CoA may arise from an abnormal formation of the desc rta. However, the exact mechanism of either type is not fully understood.

Acquired CoAs are rare, but they arise in adulthood because of severe atherosclerosis or inflammatory diseases of the aorta, such as Takayasu arteritis.

From what embryonic structure does the arch of the aorta arise?

Physiology. Because CoA is an obstructive lesion, the main hemodynamic issue is increased resistance to blood flow through the narrowed segment of the descending aorta. This is problematic because tissues distal to the coarctation require oxygen-rich blood to survive. So, the body adapts in at least one of two ways to get blood past the narrowed aorta:

- By increasing systemic blood pressure: higher blood pressures proximal to the aortic narrowing will increase arterial pressure proximal to the stenosis (eg, in the arms a ig cause of **secondary hypertension** in children.
- By developing **collateral circulations:** existing, connected minor blood vessels that send blood around an obstructed vessel generally develop slowly. Thoracic arteries commonly involved in rerouting blood past a CoA include the left subclavian, internal thoracic, axillary, and intercostal arteries (Figure 3).

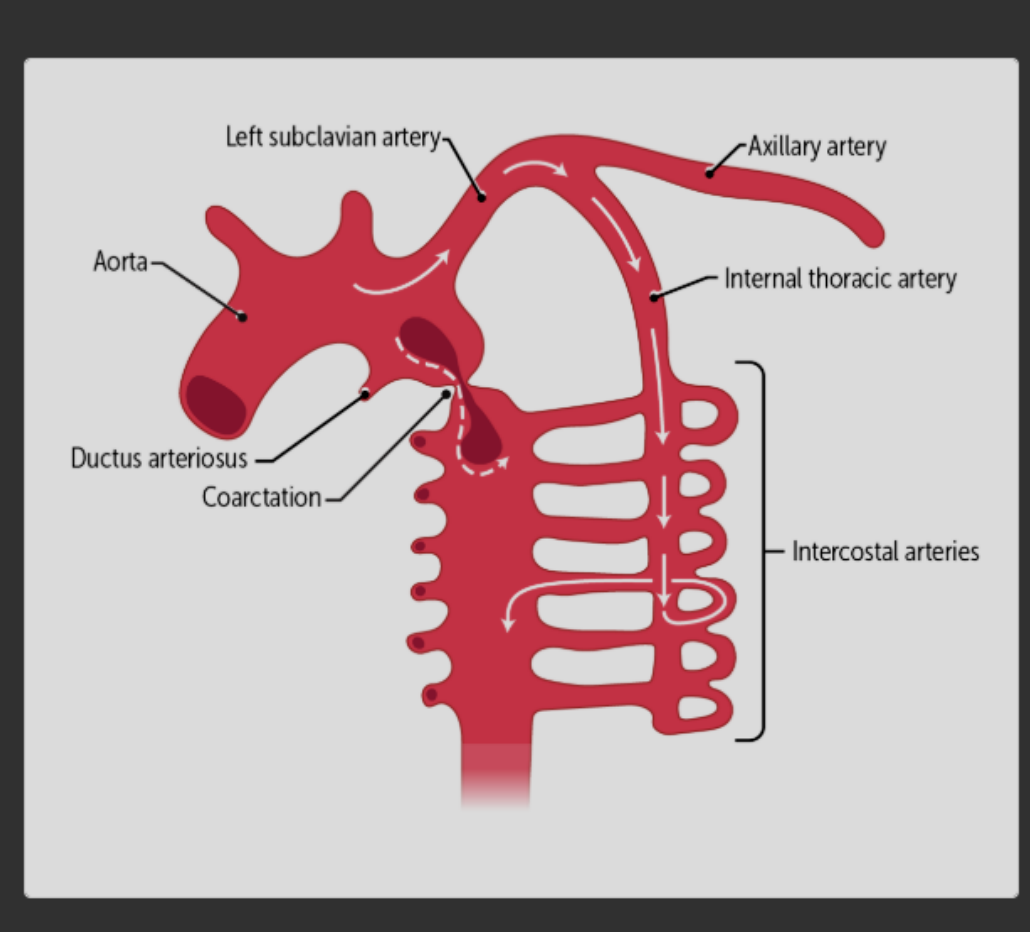


Figure 3 Collateral circulation in coarctation of the aorta

Diagnosis

CoA is diagnosed by physical exam and radiology studies.

Physical Examination. Children may have few symptoms, if any, and a normal physical exam, or they may have unexplained hypertension. Adults may present with hypertension that is easily confused with essential hypertension.

Most patients do not have physical exam findings of aortic obstruction, but, if present, some signs are helpful in diagnosis. The classic (but unusual) presenting sign of CoA is lower systolic pressure in the lower extremities than in the upper extremities. This finding presents as decreased femoral pulses or, more quantitatively, as the systolic blood pressure in the lower extremities being 15 to 20 mm Hg lower than in the upper extremities. The left upper extremity can also present with diminished pulses if the coarctation occurs proximal to the takeoff of the left subclavian artery from the aortic arch. Occasionally, a bruit can be heard in the left infraclavicular area and under the left scapula.

In a patient with more severe disease, the lower extremities may be cool to the touch because of the decreased blood flow. In neonates, the lower extremities may also appear cyanotic if the patient has a PDA in addition to the coarctation. In older children and adolescents, the elevated blood pressures can cause a midsystolic ejection murmur as blood rushes through the coarctated aorta. Flow murmurs e chest and back if large collateral arteries are present.

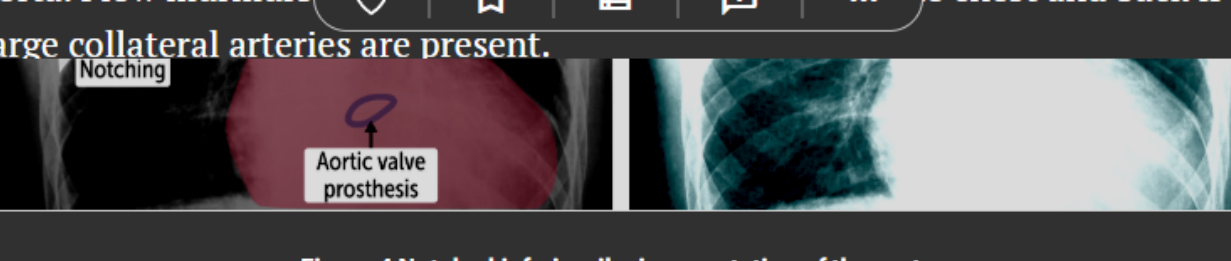
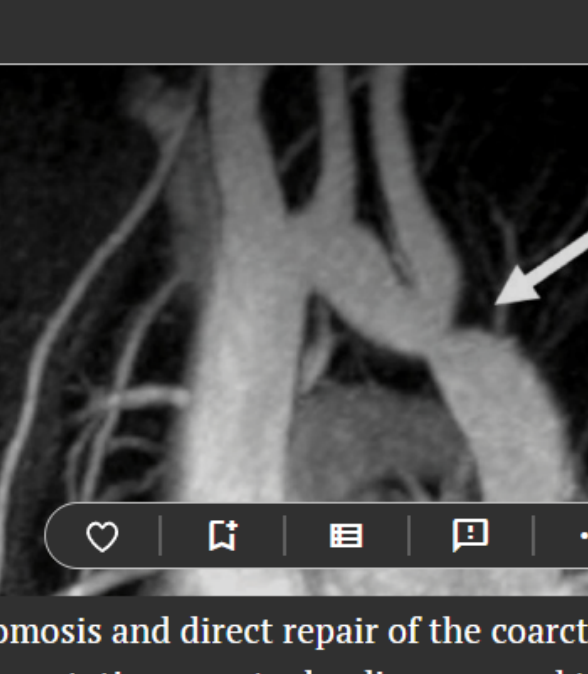


Figure 4 Notched inferior ribs in coarctation of the aorta

This phenomenon is caused by the intercostal arteries, which lie in this location and erode the lower surface of the ribs as they gradually enlarge to provide sufficient collateral circulation to the lower half of the body. Chest x-ray may also reveal an indentation of the aortic arch at the site of the coarctation.

Computed Tomography and Magnetic Resonance Imaging. Computed tomographic (CT) and magnetic resonance imaging (MRI) provide a 3D picture of the anatomic abnormalities associated with the coarctation. Although the diagnosis can be established without them, these images can be helpful in planning the appropriate intervention (Figure 5).



end-to-end anastomosis and direct repair of the coarctation. In neonates with severe preductal coarctation, prostaglandins are used to keep the ductus arteriosus open to maintain blood flow to the lower half of the body until corrective surgery can be performed.

In children, surgery is performed to prevent the development of systemic hypertension and associated complications.

Older Children and Adults. Transcatheter intervention with stenting has become the preferred approach (Figure 6).

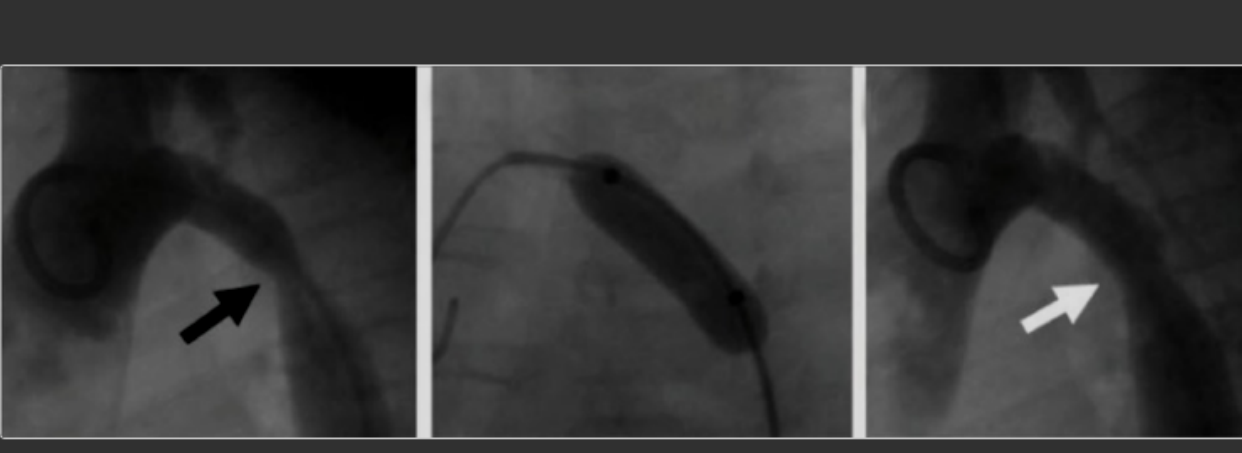


Figure 6 Catheter angiography treatment of coarctation of the aorta

Figure 6 shows a patient with CoA before treatment (left), during stenting (middle), and after placement of a percutaneous transcatheter stent (right).

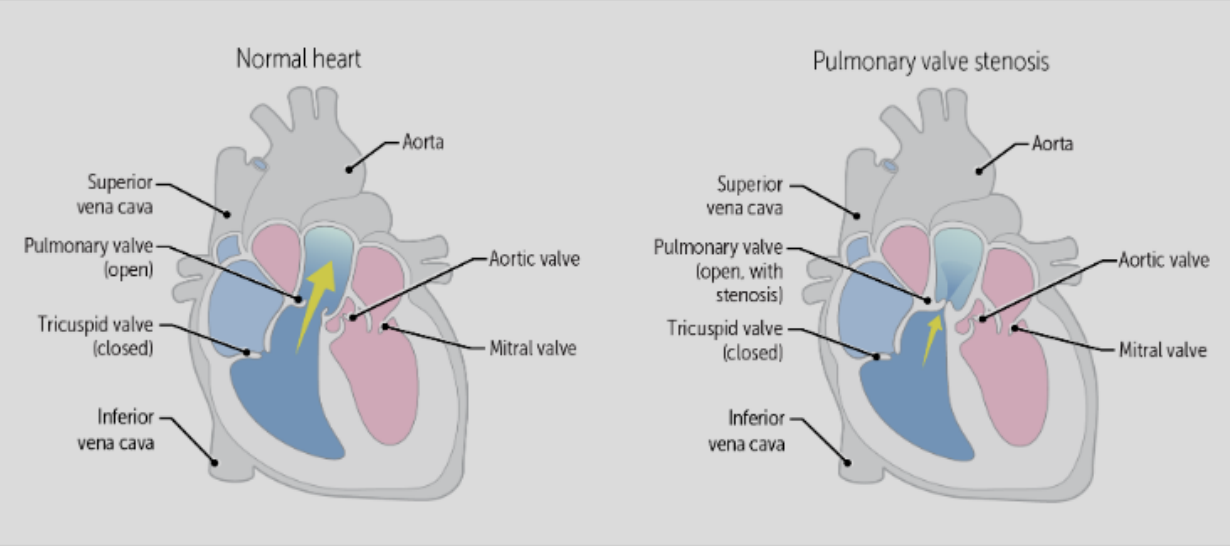


Figure 7 Normal heart vs pulmonary valve stenosis

These obstructive changes in the pulmonary valve narrow this outflow tract, decreasing the amount of blood flowing into the lungs. Although it is most common as an isolated lesion, pulmonary valve stenosis may coexist with other malformations; in fact, it is one of four coexisting malformations in tetralogy of Fallot.

Complications

Long-standing disease may lead to right ventricular hypertrophy (RVH) and then to heart failure.

What type of murmur is commonly heard in pulmonary valve stenosis?

Radiology. Echocardiography establishes the diagnosis, severity, and associated abnormalities. Chest x-ray may show cardiomegaly due to RVH.

Management

The malformed pulmonic valve is surgically replaced. Prior to replacement, stenosed valves can be managed with balloon catheter dilation, in which a catheter is advanced across the valve and a balloon inflated to dilate the narrowed valve.

What Is Congenital Aortic Valve Stenosis? (LO#1)

Congenital aortic val ely termed "bicuspid aortic valve," accounts for ~4% of CHD. In AVS, a thickened aortic

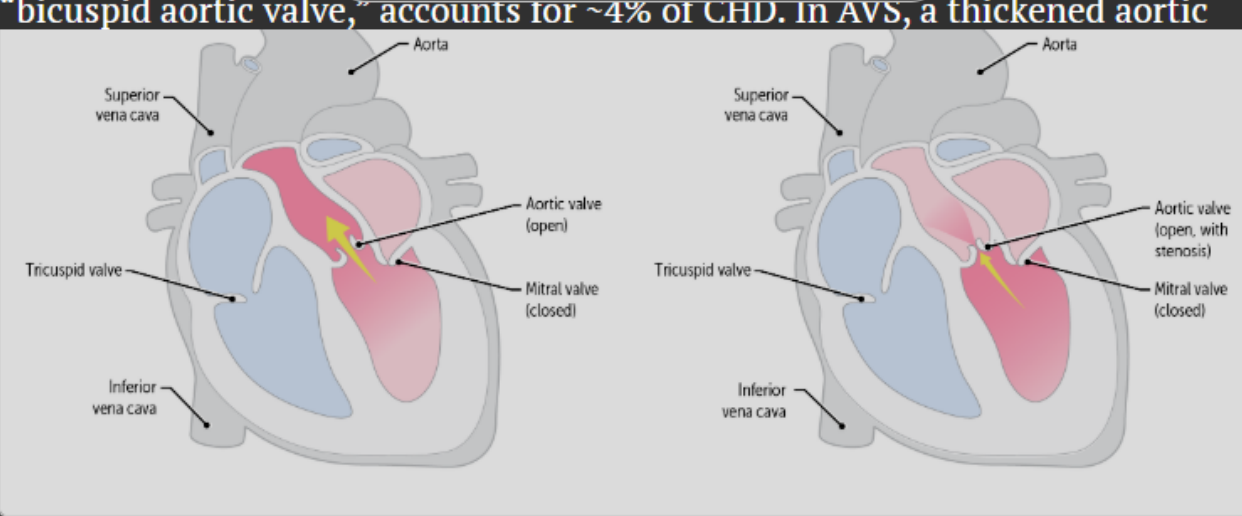


Figure 8 Normal heart vs aortic valve stenosis

In many patients with AVS, the valve itself is malformed; most often it is **bicuspid**. Having two cusps rather than the normal three predisposes the valve to fibrosis and calcification (and worsening stenosis) with age. Another form of AVS occurs when a membrane forms in the aorta itself (not the valve), causing a fixed obstruction.

Complications

AVS is a progressive disease. With increasing stenosis of the aortic valve, the LV adapts with hypertrophy, which, over time, can lead to heart failure.

Review Questions

1. **A 2-day-old girl previously diagnosed with coarctation of the aorta suddenly goes into cardiogenic shock. Closure of which of the following remnants of fetal circulation may have precipitated heart failure in this patient?**

   A. ● Ductus arteriosus
   B. ○ Ductus venosus
   C. ○ Foramen ovale
   D. ○ Umbilical artery

Hide Explanation

The correct answer is ductus arteriosus (A), an important part of fetal circulation that allows blood to bypass the pulmonary circulation during gestation. Before its closure shortly after birth, the ductus arteriosus allows blood to flow into the lower extremities in the setting of coarctation of the aorta (CoA). After it closes, blood flow to the lower extremities suddenly decreases and blood pressure dramatically increases, which can precipitate heart failure and shock. The ductus venosus (B) shunts blood from the umbilical vein directly into the inferior vena cava, and its closure does not affect symptoms in CoA. The foramen ovale (C) lies in the atrial septum, and its closu mbilical artery (D) is a vessel that takes deoxygenated blood away from the fetus to the placenta and has no impact on the symptoms of CoA.

2. **Which of the following physical exam findings is most associated with severe coarctation of the aorta?**

   A. ○ Bounding peripheral pulses
   B. ● Decreased lower extremity pulses
   C. ○ Faint pulses in the upper extremities
   D. ○ Pulsus paradoxus

Hide Explanation

The correct answer is decreased lower-extremity pulses (B). CoA impairs blood flow to the lower half of the body, leading to diminished lower extremity pulses. Bounding peripheral pulses (A) is a physical exam finding commonly associated with aortic regurgitation and truncus arteriosus. Faint pulse in the upper extremities (C) is a common physical exam finding in Takayasu arteritis. Pulsus paradoxus (D) is an abnormally large decrease in systolic blood pressure during inspiration and is commonly seen in cardiac tamponade and constrictive pericarditis.

3. **Which of the following medications can be used to prevent the rapid onset of heart failure an arctation of the aorta?**

   A. ○ Angiotensin-converting enzyme inhibitors
   B. ○ β-Blockers
   C. ○ NSAIDs
   D. ● Prostaglandins
   E. ○ Thiazide diuretics

Hide Explanation

The correct answer is prostaglandins (D), which help prevent the ductus arteriosus from closing, thus maintaining blood flow to the lower half of the body in the setting of coarctation of the aorta (CoA). Angiotensin-converting enzyme inhibitors (A) and β-blockers (B) are used to lower blood pressure but would not be useful in treating a neonate with CoA. NSAIDs (C) are more commonly used to facilitate the closing of a patent ductus arteriosus and would be contraindicated in this case. Thiazide diuretics (E) are also used to lower blood pressure.

4. **Which of the following diagnostic tests is preferred for diagnosis of pulmonary stenosis?**

   A. ○ Computerized tomography
   B. ● Echocardiography
   C. ○ Electrocardiography
   D. ○ Magnetic resonance imaging

Hide Explanation

The preferred diagnostic test for this (and most other) congenital heart disease is echocardiography (B). Computerized tomography (A), electrocardiography (C), and magnetic resonance imaging (D) may each provide helpful information but are not preferred for diagnosis.

How would you rate this Brick?

References

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Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of obstructive congenital heart defects.~~

Objective Confidence: 1/1